Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 183 (2021) 61–72

Procedia Computer Science

www.elsevier.com/locate/procedia

10th International Conference of Information and Communication Technology (ICICT-2020)

Real-time object detection based on YOLO-v2 for tiny vehicle object

Xiaohong Han*, Jun Chang, Kaiyuan Wang

Taiyuan University of Technology, No 79 yingze west street, taiyuan,030024,China

Abstract

In Automatic Driving System (ADS) and Driver Assistance System (DAS), object detection plays a vital part. Nevertheless, existing real-time detection models for tiny vehicle objects have the problems of low precision and poor performance. To solve these issues, we propose a novel real-time object detection model based on You Only Look Once Version 2 (YOLO-v2) deep learning framework for tiny vehicle objects, called Optimized You Only Look Once Version 2 (O-YOLO-v2). In the proposed model, a new structure is introduced to strengthen the feature extraction ability of the network by adding convolution layers at different locations. Meanwhile, the problem of gradient disappearance or dispersion caused by increasing network depth is solved by adding residual modules. Furthermore, in order to promote the detection accuracy of tiny vehicle objects, we combine the low-level features and high-level features of the network. The experimental findings and analysis on a KITTI dataset show that the model not only promotes the accuracy of tiny vehicle object detection but also improves the accuracy of vehicle detection (the accuracy reaches 94%) without decreasing the detection speed.

Keywords: Tiny vehicle detection; deep learning; YOLO-v2; residual module; feature fusion;

1. Introduction

In recent years, great progress has been made in object detection, particularly with the promotion of deep Convolutional Neural Networks (CNN). Considering the problems of traffic congestion and driving safety, it is valuable for the application of vehicle detection to Autonomous Driving System (ADS), Driver Assistance System

* Corresponding author. Tel: +134-8532-5973;
E-mail address: hanxiaohong@tyut.edu.cn

1877-0509
10.1016/j.procs.2021.02.031

(DAS), Intelligent Transportation System (ITS) and so on [1]. However, in complex traffic scenes, object detection still faces many challenges, such as tiny vehicles which are sheltered from each other, objects in different situations such as blurry, rainy, and night.

Hand-craft features and cascade classifier structure [2] were very popular in early object detection models. These models first get the object area in the original images by using the sliding window and then extract different types of features, finally use a classifier to detect different objects. For example, Meng et al. [3] used Haar-like features to describe the edge, linear, center, and diagonal features of face images, and then employed SVM to identify the object. These early studies have gained certain success. However, these models can only describe the appearance and shape of the objects through the different types of features and cannot express the deep feature information of the objects, which makes it difficult for the model to adapt to new datasets and makes it very difficult to detect objects in the changing traffic scene.

Driven by the theory of deep learning, object detection has made great improvement in speed and accuracy. These models are divided into two stages, in which proposals are generated in the first stage and refined in the second stage through the regional proposal strategy and Convolutional Neural Networks (CNN). For example, Zhang et al.[4] developed a model which first used binocular vision algorithm to preprocess the image, and then utilized the Faster R-CNN to detect the vehicle. Experiments showed that this model improved the speed of object detection greatly. Gao et al.[5] proposed an YOLO-based improved network structure to promote the accuracy and speed of object detection.

All the mentioned above models are trying to establish a model to satisfy the conditions of object detection in traffic scenarios. Although these models promote the accuracy and speed of object detection, due to limitations of Faster R-CNN [6], the models cannot meet real-time detection requirements. The model based on the YOLO [7] has significantly promoted the detection speed, however, due to the shallow network structure, the detection accuracy is reduced compared to the Faster R-CNN. At the same time, this model is poor in detecting tiny objects and is unable to recognize the vehicles which are sheltered from each other in traffic scenes.

Considering the problems of the above research, we propose a new Optimized You Only Look Once Version 2 (O-YOLO-v2) model based on the YOLO-v2. The highlights of this model mainly include: (1) We design the network structure by adding the convolution layer at different locations to improve ability of the network feature extraction. At the same time, we add the residual modules [8] to solve the issue, which is called gradient disappearance or dispersion mainly caused by the stacking of network layers. (2) We fuse the features extracted from different network levels to improve the detection accuracy for tiny objects. (3) The proposed model extracts deeper features while speeding up the network training simultaneously, which meets the requirements of object detection in the real-time traffic scene.

The rest of the paper is arranged as follows: Section 2 explains the theory of YOLO-v2. Section 3 introduces the technical details of the proposed O-YOLO-v2. Specifically, the overview of the put forward model is shown in Section 3.1. And the loss function of model is described in Section 3.2. In order to verify the validity of the proposed model, extensive experiments are carried out in Section 4. Section 5 summarizes the characteristics of the proposed O-YOLO-v2.

2. YOLO-v2 Principle

YOLO-v2 [9] directly predicts the bounding box and category probability by using a single neural network. The image is divided into an $S \times R$ grid of cells with that only one object can be predicted by a cell. The role of a grid cell is to detect this object whose center falls into the grid cell. The B bounding boxes, the confidence score of the B bounding boxes, and category probabilities can be predicted by every grid cell. The bounding box is composed of 5 parts: (x, y, w, h, confidence C). The (x, y) coordinates in comparison with the grid cell location point to the center of the box. As far as the range of coordinates is concerned, we normalize it to 0 to 1. The width and height of the bounding box are represented by (w, h) box dimensions, which are also created on 0 to 1 compared with the image size.

The predicted confidence scores represent two meanings: the model's confidence in object contained by the box and how accurate it believes the prediction of the box is. The confidence scores are zero shows that cell does not contain any objects. If not, we expect the confidence score is equal to the intersection over union (IOU) between the ground truth and the predicted box [10]. In some cases, a single grid cell can contain multiple objects. Figure 1 shows the computational process of the grids and bounding boxes. Figure 2 is a detection flow diagram of the YOLO-v2 algorithm. The network structure of YOLO-v2 is shown in Table 1.

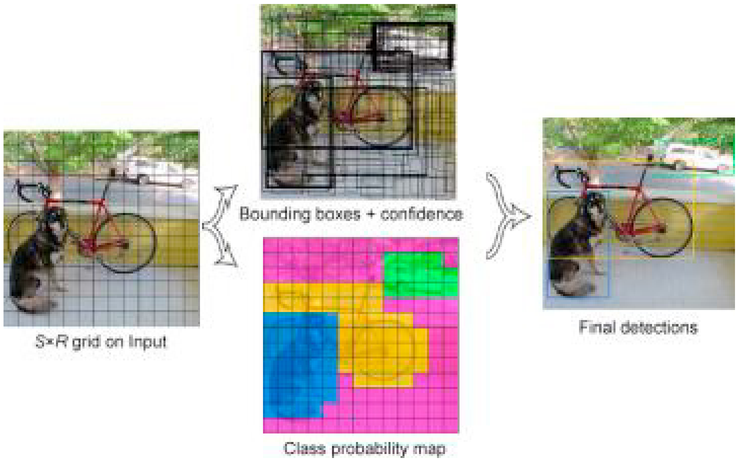

Fig. 1. YOLO-v2 model detection. First dividing the image into an S × R grid, then the model predicts B bounding boxes for each grid cell, confidence for those boxes, and C category probabilities.

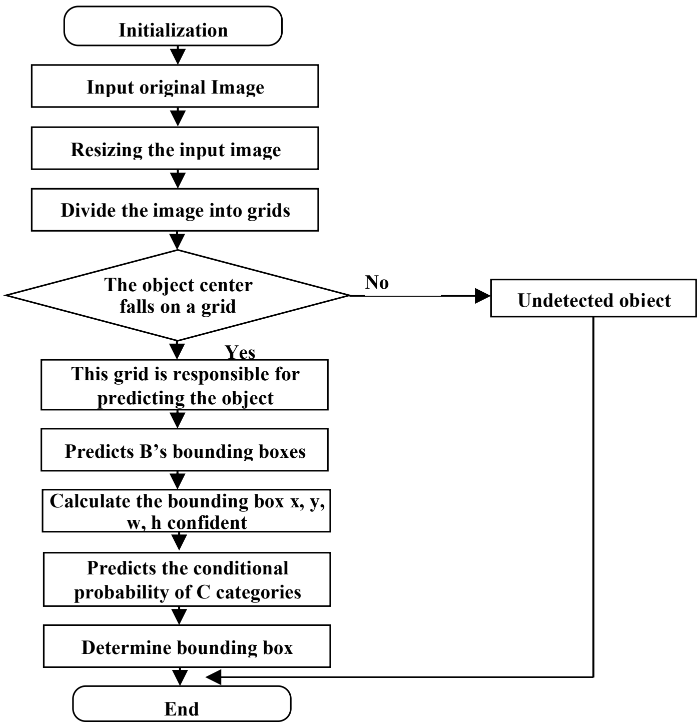

Fig. 2. YOLO-v2 detection flowchart

Table 1. Structure of YOLO-v2 network.

Number	Type	Size/Stride	Output
0	Input	-/-	416×416×3
1	Convolutional	3×3/1	416×416×32
2	Maxpool	2×2/2	208×208×32
3	Convolutional	3×3/1	208×208×64
4	Maxpool	2×2/2	104×104×64
5	Convolutional	3×3/1	104×104×128
6	Convolutional	1×1/1	104×104×64
7	Convolutional	3×3/1	104×104×128
8	Maxpool	2×2/2	52×52×128
9	Convolutional	3×3/1	52×52×256
10	Convolutional	1×1/1	52×52×128
11	Convolutional	3×3/1	52×52×256
12	Maxpool	2×2/2	26×26×256
13	Convolutional	3×3/1	26×26×512
14	Convolutional	1×1/1	26×26×256
15	Convolutional	3×3/1	26×26×512
16	Convolutional	1×1/1	26×26×256
17	Convolutional	3×3/1	26×26×512
18	Maxpool	2×2/2	13×13×512
19	Convolutional	3×3/1	13×13×1024
20	Convolutional	1×1/1	13×13×512
21	Convolutional	3×3/1	13×13×1024
22	Convolutional	1×1/1	13×13×512
23	Convolutional	3×3/1	13×13×1024
24	Convolutional	3×3/1	13×13×1024
25	Convolutional	3×3/1	13×13×1024
26	Output in 17 layer	-/-	26×26×512
27	Convolutional	1×1/1	26×26×64
28	Reorganization	-/2	13×13×256
29	Concatenated 25 and 28 layers	-/-	13×13×1280
30	Convolutional	3×3/1	13×13×1024
31	Convolutional	1×1/1	13×13×125
32	Detection		

The confidence C in YOLO-v2 is computed through Equation (1).

$$C = Pr\,(Object) * IOU \tag{1}$$

In which IOU is the joint intersection between the ground truth and the predicted box. $Pr(Object)$ is the probability of predicting whether the boundary object contains the vehicle object. If there is an object, $Pr(Object)$=1, Otherwise $Pr(Object)$=0.

The category probability p is calculated as shown in Equation (2):

$$p = Pr(Class \mid Object) * Pr(Object) * IOU \tag{2}$$

3. Proposed Detection Model

Although YOLO-v2 has the advantages of being able to directly predict the position, size, and category probability of the object from the input image and achieves good detection results, and the detection speed is fast, in real traffic scenes, we are often encountering tiny vehicle objects, which more often depend on low-level features. Due to the shallow network structure of YOLO-v2, the low-level features cannot be extracted fully through YOLO-v2 model, and the feature information is lost during the transmission process between different layers. Therefore, the multi-layer feature information is not completely used in the process of prediction, which reduces the accuracy of object location.

3.1. Proposed Detection Model

For the purpose of improving the accuracy of object detection and enhancing the detection performance of the tiny object, this paper concerns the development of a new detection model, which is called O-YOLO-v2 based on the YOLO-v2. In addition, the main objective of such an introduced new architecture is to promote the performance of feature extraction. Meanwhile, a residual module is proposed to solve the issue of gradient disappearance or dispersion, which is caused by the increased network depth through. Furthermore, in order to fully utilize the features on the shallow feature map to improve the detection accuracy of small object vehicles, the model fuses the low-level features and high-level features. Additionally, considering that the position error generated by the O-YOLO-v2 during training is greater than the classification error, we make the adjustment of the parameters of loss function to further promote the performance of model.

The proposed architecture is shown in Figure 3, in which Conv3-X represents the convolutional layer; the convolution kernel size is 3×3; the channel number is X; the default step size is set to 1; Pool2 represents the maximum pooling layer; the filter is 2×2; the default step is set to 2; Residual represents the residual block structure. The architecture contains 31 convolutional layers, 6 pooling layers, 1 feature reorganization structure and 11 residual modules. The input to the network is 416 × 1248 × 3 and the output is 13 × 39× 30. As shown in Figure 3, two residual modules are constructed through the 6th, 7th, 10th, and 11th convolutional layers (the 6th and the 7th for the first residual module; the 10th and the 11th for the second residual module). After the 11th layer, a new residual module is added which consists of two layers of convolution. After that, two more residual modules are established through the 16th, 17th, 18th, and 19th convolutional layers (the 16th and the 17th for the first residual module; the 18th and the 19th for the second residual module). After the 19th layer, two new residual modules are added. Then, another two residual modules are constructed through the 26th, 27th, 28th, and 29th convolutional layers (the 26th and the 27th for the first residual module; the 28th and the 29th for the second residual module). After the 29th layer, two convolutional layers are moved and two new residual modules are added. Finally, the feature maps outputted from the 13th ,23rd, and 33rd layer are fused.

As shown in Figure 3, the feature map (52 × 156 × 256) outputted from the 13th layer, the feature map (26 × 78 × 512) outputted from the 23rd layer, and the feature map (13 × 39 × 1024) outputted from the 33rd layer are fused together. The feature map (52 × 156 × 256) outputs the feature map (13 × 39 × 256) through an average pooling. The feature map (13 × 39 × 256) is outputted from the feature map (26 × 78 × 512) after feature reorganization and a convolutional layer. After the feature maps of the 13th, 23rd, and 33rd layers are added, the feature map (13×39×1536) is outputted and then is inputted to the next layer.

Because the features of tiny objects are mainly contained in the shallow layers of the network and the features of large objects are contained in the deep layers of the network, we fuse multi-layer features of the low-level features and high-level features to get the enhanced feature expression ability of tiny objects which can get the higher accuracy of tiny object detection.

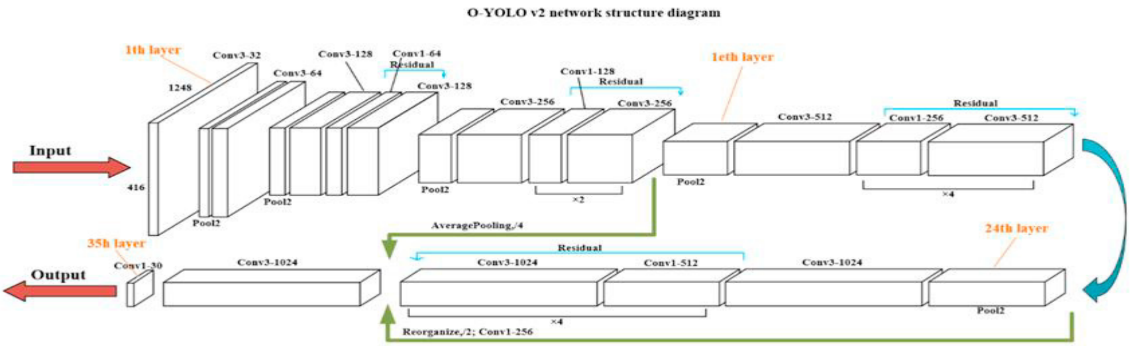

Fig. 3. O-YOLO-v2 network structure diagram

To solve the problem of gradient disappearance or dispersion, which is caused by the increased network depth, this paper proposes to add residual modules into the model. Besides, the added residual modules can also enhance the feature extraction ability of the model. The proposed residual module is shown in Figure 4, which consists of a convolutional layer of 1×1, a convolutional layer of 3×3, and an identity mapping. The convolutional layer of 1 × 1 and the convolutional layer of 3 × 3 are filled with the SAME mode. Under the circumstance of using the SAME mode of filling, the size of feature map keeps unchanged and is equivalent to the size of identity mapping, so they can be added directly.

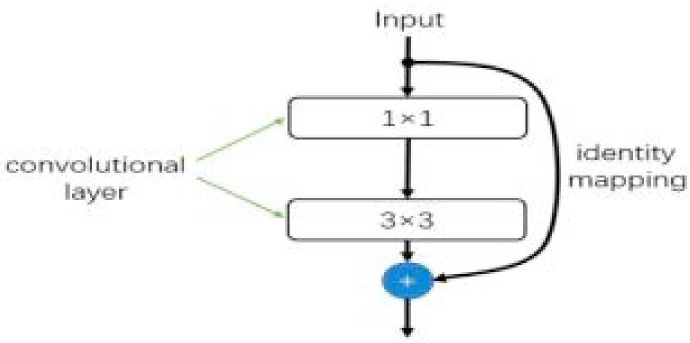

Fig. 4. Residual module

3.2. Loss Function

Considering that the position error generated by the proposed O-YOLO-v2 during training is greater than the classification error, we make the adjustment of parameters of the loss function. We increase the weight of position error to 5.5 (the original position error weight is 5), and reduce the weight of classification error to 0.5 (the original confidence weight is 1). In addition, we use the cross-entropy loss function to calculate the classification error, which accelerates the convergence process compared to the original square-sum loss function. The loss function used in this paper is calculated as shown in Equation (3), (4), and (5):

$$L_{object} = 5.5\sum_{i=0}^{S*R}\sum_{j=0}^{B}[(\hat{x}_i - x_i)^2 + (\hat{y}_i - y_i)^2] + 5.5\sum_{i=0}^{S*R}\sum_{j=0}^{B}\left[\left(\sqrt{\hat{w}_i} - \sqrt{w_i}\right)^2 + \left(\sqrt{\hat{h}_i} - \sqrt{h_i}\right)^2\right] + 0.5\sum_{i=0}^{S*R}\sum_{j=0}^{B}(\hat{c}_i - c_i)^2 - \sum_{i=0}^{S*R}\sum_{class=car}(\hat{p}_i * \log(p_i)) \tag{3}$$

$$L_{no-object} = 0.5\sum_{i=0}^{S*R}\sum_{j=0}^{B}(\hat{c}_i - \hat{c})^2 \tag{4}$$

$$L_{total} = L_{object} + L_{no-object} \tag{5}$$

In the Equation (3), L_{object} indicates the loss of model of the object falling into bounding box; In the Equation (4), $L_{no\text{-}object}$ indicates the loss of the model of the object not falling into the bounding box; In the Equation (5), L_{total} indicates the total loss of the model; x_i, y_i, w_i, h_i, c_i, p_i are true label values; $\hat{x}_i$, $\hat{y}_i$, $\hat{w}_i$, h_i, $\hat{c}_i$, $\hat{p}_i$ are model prediction values.

4. Experiments

For the purpose of verifying the performance of the proposed O-YOLO-v2, we compare it with the YOLO-v2 and the YOLO-v3.

4.1. Experimental Setup

We adopt the TensorFlow and Keras framework, using the Python 3 programming language to train our deep learning model. The model training was performed using Nvidia Tesla K40, 11G memory, Centos7.0 operating system, 24G memory. The model test was performed using Nvidia GeForce 1050, 4G memory, Intel(R) Core(TM) i7-7700HQ CPU@2.8GHz, 16G memory, Windows10 operating system.

4.2. Datasets

We use the KITTI dataset for model training and testing. KITTI is a public dataset used for verifying detection models including vehicle detection, vehicle tracking, and semantic segmentation. We select the 2d object images in the KITTI dataset as our dataset. There are a total of 7481 real traffic scene images in the KITTI dataset, in which 6178 images are randomly selected as the training set and the remaining 1303 images as the verification set. In order to make the images fit to our network structure and not lose information during the convolution process, the interpolation method is used to interpolate on the right and below parts of the images and change the image size to 416×1248.

There are eight categories in the KITTI dataset. We change the category labels of "Van", "Truck", and "Tram" to "Car" category and the original "Car" category remains unchanged. The other categories are ignored as background. According to the directory structure of VOC [11] dataset, we convert the annotation file of the dataset into an XML format. The left side of Figure 5 is an example of original images and the right side of Figure 5 is an example of the processed images.

Fig. 5. The left side is the original image of the KITTI dataset; the right side is the image (416×1248) after interpolation.

4.3. Evaluation Index

IOU, *Recall*, *Precision*, and *Average Precision* (*AP*) are used to evaluate the O-YOLO-v2.

IOU is the intersection over union (*IOU*) between the ground truth and predicted bounding boxes generated by the model, which is shown in Equation (6), in which B_{pb} is the predicted bounding box; B_{gt} is the ground bounding box; *area*() is the area of the union shape or intersection shape of B_{pb} and B_{gt}. When the condition that *IOU* is greater than the threshold value T_{IOU} is met, the bounding box is deemed as the correct result. The correlation between ground truth and prediction value is measured based on this principle; the correlation and the value obey a proportional relationship.

$$IOU = \frac{area(B_{pb} \cap B_{gt})}{area(B_{pb} \cup B_{gt})} \geq T_{IOU} \tag{6}$$

The AP of our detection model is calculated by *IOU*. The predicted bounding box of B_{pb} is obtained by using the model to predict the input image. If *IOU* of B_{pb} and B_{gt} exceeds the threshold value T_{IOU}, and the variables obey the law of Equation (6) at this point, it is considered to be a correct prediction.

Figure 6 shows an instance of detecting a vehicle in an image. Red or blue box denotes the predicted bounding box while the green box signifies the ground truth bounding box. Our objective is to compute the *IOU* between predicted bounding boxes and ground truth. When the condition that *IOU* exceeds the 50% threshold value is met, the experimental result is defined as true positive (*TP*), and false positive (*FP*) represents the value less than threshold. The *FN* illustrates that the image does not contains any vehicles, but as a matter of fact there is a vehicle in the image.

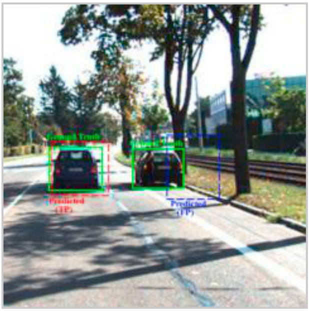

Fig. 6. Example of calculate intersection over union (IOU)

The accuracy of the object detector is commonly evaluated by *IOU*. The significance of *IOU* can be illustrated as follows: assigning anchor boxes during preparing training datasets; the usefulness using non-max suppression algorithm to clean up that model predicts multiple boxes for the same object. The T_{IOU} is set to be 0.5, which means the same region is covered by at least half of the ground truth and the predicted box. When *IOU* exceeds 50% threshold value, the vehicle is successfully predicted.

Precision is defined as the true positive divided by the identified image, which is calculated as Equation (7), in which *TP* represents the True Positive; *FP* denotes the False Positive.

$$Precision = \frac{TP}{TP+FP} \tag{7}$$

Recall is defined to be a ratio of relevant objects retrieved and the quantity of all relevant objects in the images, which is calculated as Equation (8), in which *FN* is False Negative.

$$Recall = \frac{TP}{TP+FN} \tag{8}$$

The denominator of *Recall* is defined to be true positives plus false negatives. In Equation (8), the sum of these two values is equal to the total number of vehicles of ground truth.

The last evaluation method, *AP*, can be represented by the area under the Recall and Precision curves. Experiments show that the value of *AP* is directly proportional to the detection accuracy. By calculating the values of *TP*, *FP*, and *FN* of each sample, the *Precision* and *Recall* values corresponding to each sample are obtained, then the Recall and Precision curves of *precision* and *recall* is obtained. The *AP* value, which falls 0 to 1, is calculated as Equation (9), in which f_{PR} represents the Recall and Precision curves.

$$AP = \int f_{PR}\, dr \tag{9}$$

4.4. Training

We train the O-YOLO-v2 model, the YOLO-v2 model and the YOLO-v3 model on the KITTI dataset to compare their performance.

For the purpose of speeding up the training speed of model, we chose the Adam momentum optimizer. In this experiment, the impulse constant is defined to be 0.9; the weight attenuation coefficient is defined as 0; and we adjust the learning rate to 0.00005. We set batch-size as 4, that is, 4 samples are processed for each iteration on the training set and 240 epochs are performed. The total iterative number of training is 312,720. Using the processed KITTI dataset, obtaining the weights of final vehicle detection model approximately took 4 days on the GPU.

Figure 7 shows the loss reduction curve during the model training, in which the loss value of O-YOLO-v2 is reduced to 0.027, the loss value of YOLO-v2 is reduced to 0.032, and the YOLO-v3 is reduced to 0.030.

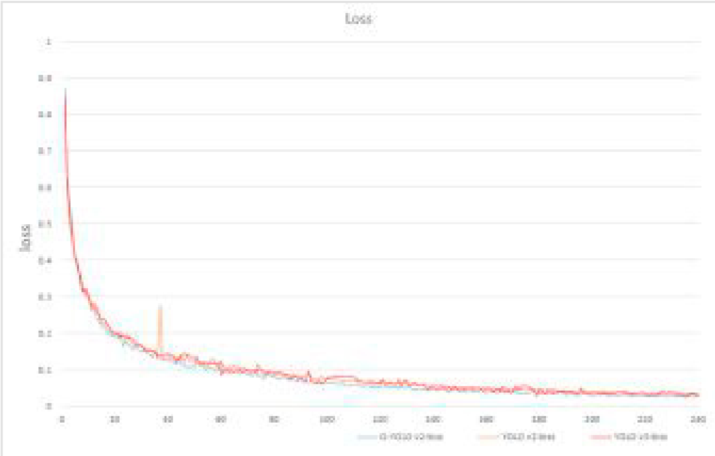

Fig. 7. Loss reduction curve during the network training process

4.5. Experiments and Analysis

As is shown in Table 2, different experimental results represent performance of the proposed O-YOLO-v2 model, YOLO-v2 model, and YOLO-v3 model on the test set. The conclusion that the proposed O-YOLO-v2 model maintains the same detection speed as the YOLO-v2 model may be depicted as in Table 2. However, Recall increased by 6%, Precision increased by 3%, and AP increased by 7%. Compared with the YOLO-v3 model, Precision of the proposed O-YOLO-v2 model increased by 1%. Compared with the YOLO-v3, although Recall and AP of the proposed O-YOLO-v2 reduced by 2% and 1% respectively, the detection speed of O-YOLO-v2 has made great progress which has increased by 53%.

In summary, the proposed O-YOLO-v2 improves the detection accuracy while maintaining the detection speed compared with YOLO-v2. Compared with YOLO-v3, although the proposed O-YOLO-v2 sacrifices detection accuracy to some extent, the detection speed improves greatly.

Table 2. Performance comparison of O-YOLO v2 model, YOLO-v2 model, and YOLO-v3 model on test set

Networks	AP	Recall	Precision	Speed/s
O-YOLO-v2	0.94	0.94	0.94	0.17
YOLO-v2	0.87	0.88	0.91	0.17
YOLO-v3	0.95	0.96	0.93	0.36

Figure 8 shows the Recall and Precision curves of the proposed O-YOLO-v2 model, the YOLO-v2 model, and the YOLO-v3 model on the test set. As shown in Figure 8, the blue curve is the Recall and Precision curves of the O-YOLO-v2 model on the test set. The green and red curves represent the Recall and Precision curves of the YOLO-v2 model and the YOLO-v3 model respectively on the test set.

It can be seen that the red curve is closer to the (1,1) coordinate point and the area formed through x axes, y axes, and the red Recall and Precision curves curve is larger, thus has a higher AP value. Compared with the red curve of YOLO-v3, the AP of the O-YOLO-v2 reduced by 1% on the test set.

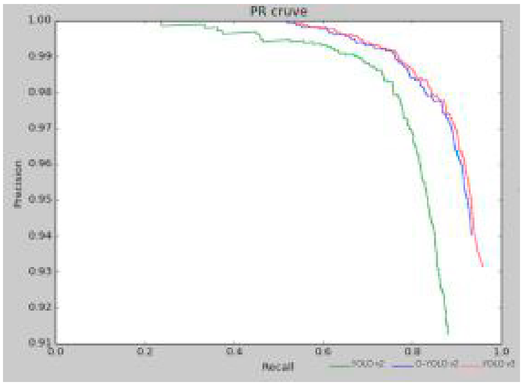

Fig. 8. Comparison of the Recall and Precision curves of the O-YOLO-v2 model, the YOLO-v2 model, and the YOLO-v3 model on the test set

Figure 9 is the comparison results of the YOLO-v2 model and O-YOLO-v2 model on the test set. From Figure 9 we can see that the proposed O-YOLO-v2 model can detect more vehicle objects and also has good detection results for distant tiny object vehicles.

Fig. 9. The results of the two models on the test set, the left side is the test results of YOLO-v2 model, the right side is the test results of O-YOLO-v2 model.

As is shown above, under the condition without decreasing the detection speed, the proposed model improves the *precision* and *recall* of model in object detection. Compared with the other two models, the proposed O-YOLO-v2 model can be more efficient for tiny vehicle object detection in real-world scenarios.

5. Conclusion

In this paper, we propose a novel object detection model called O-YOLO-v2 based on the well-known YOLO-v2. In this mode, the model, based on a new network architecture, has a better detection performance than YOLO-v2 for tiny object detection. The introduced architecture enhances the feature extraction ability of network, and at the same time, the residual modules are added to solve the problem of gradient disappearance or dispersion, which is mainly caused by the stacking of network layers. Additionally, under the condition of introducing multi-dimensional features confusion, the proposed model has improved the detection accuracy of tiny objects. As the results of experiment illustrate, the O-YOLO-v2 model obtains higher detection accuracy and basically meets the requirements of video detection. It can not only be used for vehicle object detection in real traffic scenarios but also for object detection in other fields.

Although the model proposed in this paper has achieved better detection results, there is room for improvement in model accuracy. At the same time, the images used in this paper are all under good illumination conditions.

Therefore, the network model needs to be further verified for the scenes with insufficient lighting conditions, which will be further studied in the future.

Acknowledgements

Research Project Supported by the Natural Science Foundation of Shanxi Province, China (Grant No. 201801D121136); Research Project Supported by Shanxi Scholarship Council of China (Grant No. HGKY2019024); the Nation Natural Science Foundation of China (Grants No: 61872260 and 61772358).

Compliance with ethical standards Acknowledgments

Conflict of interest: All the authors in this study declare that they have no conflict of interest.

Ethical approval: This article does not contain any studies with human participants or animals performed by any of the authors.

References

1. LIN Fuchun, ZHANG Rongfen, & LIU Yuhong. (2018). Design of Intelligent Auxiliary Driving System Based on Deep Learning. Journal of Guizhou University(Natural Science), 35(1), 73-77.
2. Liao, S., Jain, A. K., & Li, S. Z. (2016). A fast and accurate unconstrained face detector. IEEE transactions on pattern analysis and machine intelligence, 38(2), 211-223.
3. Meng Qingtao. (2008). Face Detection Based on Haar-Feature Probability Distribution and SVM. (Doctoral dissertation, East China Normal University).
4. Zhang Qi, Hu Guangdi, Li Yusheng, & Zhao Xin. (2018). Binocular vision vehicle detection method on improved Fast-RCNN. Journal of Applied Optics, 39(06), 75-81.
5. GAO Zong, LI Shaobo, CHEN Jinan, & LI Zhengjie. (2018). Pedestrian Detection Method Based on YOLO Network. Computer Engineering, 44(5), 215-219+226.
6. Ren, S., He, K., Girshick, R., & Sun, J. (2015). Faster r-cnn: towards real-time object detection with region proposal networks. IEEE Transactions on Pattern Analysis & Machine Intelligence, 39(6), 1137-1149..
7. Redmon, J. , & Farhadi, A. . (2017). [ieee 2017 ieee conference on computer vision and pattern recognition (cvpr) - honolulu, hi (2017.7.21-2017.7.26)] 2017 ieee conference on computer vision and pattern recognition (cvpr) - yolo9000: better, faster, stronger. 6517-6525.
8. He, K. , Zhang, X. , Ren, S. , & Sun, J. . (2015). Deep residual learning for image recognition.
9. Redmon, J.; Divvala, S.; Girshick, R.; Farhadi, A. You Only Look Once: Unified, Real-Time Object Detection. In Proceedings of the 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Las Vegas, NV, USA, 27–30 June 2016; pp. 779–788.
10. Redmon, J.; Farhadi, A. YOLO9000: Better, Faster, Stronger. In Proceedings of the 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Honolulu, HI, USA, 21–26 July 2017; pp. 6517–6525.
11. Everingham, M. , & Winn, J. . (2006). The pascal visual object classes challenge 2007 (voc2007) development kit. International Journal Computer Vision, 111(1), 98-136.